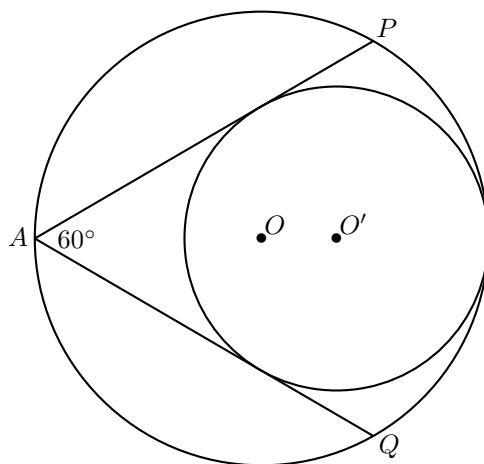


Trøndelagkonkurransen

Variety of Problems

1. The figure below shows two circumferences of centers O and O' tangent to each other. A in the larger circumference is collinear with O and O' . The chords AP and AQ are tangent to the inner circumference. If $\angle QAP = 60^\circ$, determine the ratio of the areas of the circles.



2. Find all real numbers a and b such that

$$\begin{aligned}a + b &= 14, \\ a^3 + b^3 &= 812.\end{aligned}$$